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Paleomagnetism of Cretaceous and Paleogene sedimentary rocks from the Kopet-Dagh and its tectonic implications

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Abstract

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In the Kopet-Dagh Mountains (the northernmost part of the Alpine fold belt east of the Caspian Sea), about 600 oriented samples from 21 sites were taken from the Middle Cretaceous through Oligocene sedimentary series. Prefolding and most probably primary remanence were isolated in Middle Cretaceous terrigenous rocks and Upper Cretaceous–Paleocene limestones and marls after stepwise thermal cleaning; only recent postfolding magnetization was found in Eocene–Oligocene clays. The Middle Cretaceous mean pole (75° N, 153° E) agrees well with that from extra-Alpine Eurasia. A strong negative correlation between paleomagnetic inclinations and clay content in the Upper Cretaceous–Paleocene sedimentary rocks is interpreted as arising from diagenetic compaction. The limestones least affected by this process also yielded a paleopole (78° N, 179° E) closely matching the Eurasian data. The bends of fold axes in the Kopet-Dagh are primary; tectonic rotations are discovered only within the local wrench zone in the southwestern part of the region. The close agreement of paleopoles from the Kopet-Dagh and extra-Alpine Eurasia suggests that these land masses have been welded together since the Middle Cretaceous.

Introduction

During the last two decades, tectonic movements of different type and scale have been paleomagnetically detected within fold belts all over the world. A large number of such investigations have been carried out within the Alpine belt, but even there the data distribution both in space and time is not at all uniform. Besides, most sampling localities are situated well within folded areas, and thus little is known about the tectonic history of the outermost parts of the belt.

The Kopet-Dagh Mountains are situated in the northernmost part of the Alpine belt, between the Turan plate and ranges of northern Iran. Paleomagnetic data are numerous for Plio–Pleistocene rocks (Trubikhin, 1973), but they are very scarce for older rocks (Mamedov and Nazarov, 1972; Gurary, 1977). Moreover, the objective of these works was magneto–stratigraphy, and attention was mainly paid to lateral tracing of paleomagnetic zones and not to accurate determination of paleomagnetic directions.

The author has studied Cretaceous and Paleogene rocks in the Kopet-Dagh in order to obtain tectonically meaningful data. Some of these data have already been published (Bazhenov, 1979, 1983); here the complete data set is presented.

Geological setting and sampling

The Kopet-Dagh are the northernmost part of the Turkmeno–Khorasanian Mountains, which stretch over 1000 km between the Alborz Range in the west and the Parapamisus in the east. This folded zone is bounded by the relatively rigid

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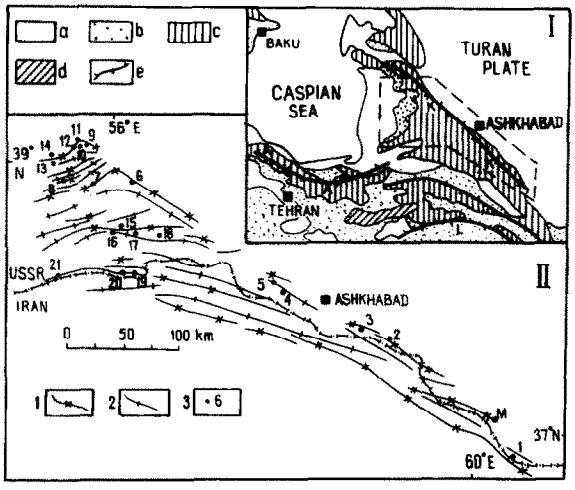


Fig. 1. Schematic geological map (I) and the structural pattern and distribution of sampling sites (II) of the Kopet-Dagh region. I. Legend: *A*—Alborz range; *K*—Kopet-Dagh; *L*—the Lut block; *a* = Neogene and Pleistocene rocks; *b* = Paleogene rocks; *c* = Mesozoic rocks; *d* = older formations; *e* = major faults. II. Legend: *1*, *2* = anticlinal and synclinal folds, respectively; *3* = sampling sites; *M*—sampling site from Mamedov and Nazarov (1972).

blocks of Central Iran in the south and the Kopet-Dagh Foredeep in the north. The sampling region is the approximately 600-km-long part of the Kopet-Dagh within the limits of the USSR (Fig. 1). There is an exposed, abundantly fossiliferous miogeosynclinal sedimentary series more than 3 km thick. The Upper Jurassic limestones outcrop mainly within small fault-bounded blocks. The more widespread Neocomian carbonate rocks are successively overlain by the Middle Cretaceous terrigenous sediments, and limestones and marls of Late Cretaceous age. In the Western and Central Kopet-Dagh, limestones disappear almost completely by the end of the Mesozoic, and clayish marls of Paleocene age are overlain by marine Eocene–Oligocene clays. White, light-grey and greenish-grey rocks prevail. Thin layers of redbeds are present only in the Early Senonian and Paleocene parts of the section. In the Eastern Kopet-Dagh, light-colored limestones, dolomites and marls alternating with grey and red sandstones and mudstones overlay the Upper Cretaceous carbonates. These Uppermost Cretaceous–Lower Paleocene sediments have a thickness of about 200 m.

No angular unconformities were found within the Cretaceous–Paleogene sequence in this region, but there are some hiatuses, sometimes with traces of underwater erosion, e.g., in the Coniacian (Atabekyan and Likhacheva, 1961). All of these sediments are unmetamorphosed and some marls and clays disintegrate quickly when exposed to water. No magmatic rocks are known here.

Folding in the Kopet-Dagh took place in the Neogene, the age of deformation being progressively younger from south to north. The structural pattern is comparatively simple, and, owing to generally good exposures, has been investigated in detail (Rastsvetaev, 1972). The strike of fold axes changes gradually from NNW in the east to NNE or NE in the west. Folds are cut by numerous low-amplitude faults of different kinds (not shown in Fig. 1).

The Middle–Upper Cretaceous and Paleogene rocks were sampled at a number of sites distributed all over the Kopet-Dagh (Fig. 1). When possible, both limbs of folds were studied. One hand sample oriented with magnetic compass was taken from a layer. The author tried to study all lithologic units, but some proved to be too friable for sampling (for sampling details see Table 1). Thus, the whole collection of about 600 samples reflects qualitatively, but not quantitatively, the lithology of the sequence studied.

Methods and procedures

Three to five specimens were cut from each hand sample in the laboratory. Two specimens per sample were subjected to stepwise thermal cleaning up to 400 °C in a furnace shielded with two μ-metal layers and placed within a three-pair-system of Helmholts coils. If the within-sample scatter was high, one or two specimens were added. The mutual orientation of specimens was changed after each heating step in order to detect and evaluate any laboratory-induced magnetizations. All measurements of natural remanent magnetization (NRM) or its components were made with Czechoslovakian JR-3 and JR-4 spinner magnetometers. The isothermal remanent magnetization (IRM) acquisition and the IRM thermal demagnetization curves were obtained for some representa-

TABLE 1

Sampling details *

Site	Age	T [d]	n	$A°$	$d°$	Lithology
1(1)	m–Tp	120	40	58	44	L, M, Rb
2(2)	Te	40	14	35	65	C
3(3)	al–Tp	1000	28	196	36	S, L, M
4(4) [b]	al	200	13	–	–	S
5a	al–cp	600	14	359	42	S, L, M
5b	al–cp	700	17	165	10	S, L, M
6(5, 1)	ap–m	1100	70	215	47	S, L, M, Rb
7a(6)	st–cp	300	11	145	66	L, M
7b(6)	st–cp	200	9	319	27	L, M
8a(7)	Te–o	250	9	335	60	C
8b(7)	Te–o	300	12	161	85	C
9(8, 2)	c–Te	1000	59	190	45	S, L, M, C, Rb
10(8, 3)	cn–To	1000	46	135	38	L, M, Rb
11(10, 4)	cn–Te	900	46	317	57	L, M, C, Rb
12a	al	100	8	153	12	S
12b	al	300	15	308	41	S
13(11)	al–Tp	900	25	165	75	S, L, M, C
14(12)	al–c	250	13	357	30	S
15(–, 5)	cp–Tp	300	18	335	30	L, M
16(–, 6)	cp–Tp	400	20	190	40	L, M
17	al–m	900	30	174	36	S, L, M
18a	al	130	11	165	37	S
18b	al	100	9	5	40	S
19	al–c	250	25	346	55	S
20a	al–c	200	13	345	70	S
20b	al	40	11	187	27	S
21	al–c	200	18	358	38	S

* Site: Numbers in brackets correspond to site numbers in Bazhenov (1979) and Bazhenov (1983), respectively. ([b] Bedding attitudes are variable here.) Age is given up to stage, when possible: ap = Aptian, al = Albian, c = Cenomanian, t = Turonian, st = Santonian, cn = Coniacian, cp = Companian, m = Maastrichtian. T = true thickness sampled in meters; n = number of samples; $A°$ = mean dip direction; $d°$ = mean angle of dip. Lithology: S = sandstones, L = limestones, M = marls, C = clays, Rb = red beds.

tive samples. These measurements were made with an astatic magnetometer.

Results of the stepwise thermal cleaning were analyzed with stereonets. If laboratory-induced magnetizations were negligible and magnetization intensities were well above the noise level of magnetometers, the paleomagnetic directions after the final heating were always preferred. Otherwise, one of the previous steps was used. Owing to furnace design, samples could be heated only up to 400°C, and the reliability of paleomagnetic results was proved mainly by field tests.

When possible, site means were used for fold tests (Watson, 1966; McFadden and Jones, 1981). First, this test was applied on a local scale, that is, to results from two limbs of the same fold, and then it was used on regional scale. If it was positive, the corresponding site mean or regional mean directions were treated as the true directions of prefolding magnetization. If this test was negative, the converging remagnetization circles (the CRC method) was applied to separate NRM components (Halls, 1976). The same method was also applied to sets of site means irrespective of fold test results in order to determine postfolding and prefolding components independently. The component directions and their confidence limits were computed according to Halls (1976) and Mardia (1972), respectively. The concentration parameter ratio (CPR) test (McElhinny, 1964) was used only

for sites with very variable bedding attitudes. It has been shown recently (McFadden and Jones, 1981) that some basic assumptions of the CPR test are invalid. In any case, the increase in the concentration parameter after tilt correction does not necessarily mean the coincidence of the tilt-corrected paleomagnetic vector with the prefolding direction when the magnetizations contain two or more components. Last, when both normally (N) and reversely (R) magnetized samples were found at the same site, the reversal test was applied to such data.

Results

Middle Cretaceous terrigenous rocks

These sediments, mostly sandstones, are characterized by NRM intensities of about $1 \cdot 10^{-6}$ emu/cm^3. Rock-magnetic studies have revealed that magnetite predominates in this collection, though a minor amount of haematite was detected in a very limited number of samples (Fig. 2). Usually, thermal demagnetization at 200° C left 20–30% of the original remanence. Afterwards, magnetization intensity decreased slower, but cleaning was often terminated well before 400° C because of the magnetization dropping below the noise level of the magnetometer. The within-sample scatter of paleomagnetic directions varied widely, the concentration parameter values ranging from 1 to more than 1000. Some samples were rejected if the specimen vectors did not meet the randomness criteria set forth by Irving (1968), where for $n = 3$; $r < 2.62$ and for $n = 4$; $r < 3.10$ (n is the number of specimens cut from the sample, r is the length of the vector resultant).

In the course of thermal demagnetization, most site means shifted along great circle paths (Fig. 3). Directional changes mainly took place below 200° C and afterwards they were much less pronounced or absent (within error limits). An intersection of these remagnetization circles defines both post-folding and prefolding NRM components before and after tilt correction, respectively

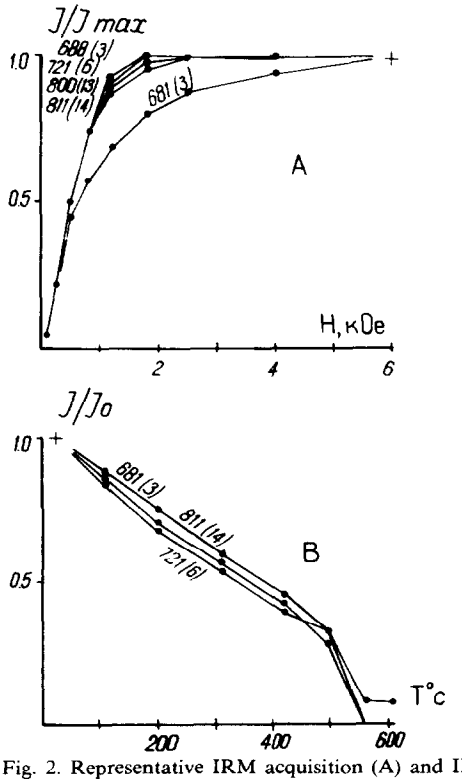


Fig. 2. Representative IRM acquisition (A) and IRM thermal demagnetization curves (B) for Middle Cretaceous rocks. Samples (sites) are numbered.

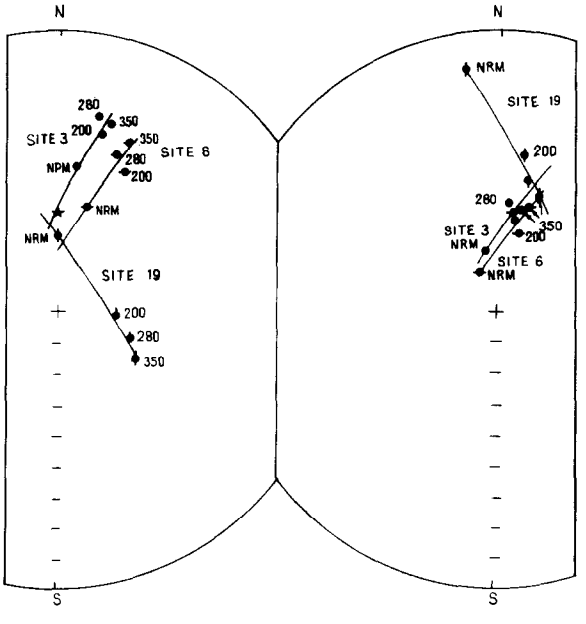


Fig. 3. Representative site remagnetization circles for Middle Cretaceous data before (left) and after (right) tilt correction. Successive site-mean directions are shown after thermal cleaning. The star is the modern dipole field direction. The solid (open) symbols denote downward (upward)-pointing vectors (The same convention is preserved everywhere).

TABLE 2

Middle Cretaceous paleomagnetic results *

Site	Age	n		$D°$	$I°$	K	α_{95}	*Plat* (°N)	*Plong* (°E)
		a	b						
3	al–c	15	12	6	54	38	7.1	84	183
4	al	13	11	5	58	20	9.4	86	137
5	al–c	14	9	17	58	–	–	77	140
6	ap–c	19	15	19	57	33	6.8	75	147
9 [a]	c	14	8(24)	330	65	4	14.6	356	67
12	al–c	23	18(54)	15	54	4	10.4	77	161
13	al–c	9	6	34	60	42	8.1	64	130
14	al–c	13	10	30	54	24	9.1	66	147
17 [b]	al–c	11	5	65	58	21	13.8	40	124
18 [b]	al	20	16(34)	60	59	–	–	44	125
19	al–c	24	15	18	51	29	6.7	74	164
21	al–c	18	14	21	49	24	7.7	71	164
Overall mean			$N = 8$	19	55	160	3.9	75	153
(Class A data, sites 3, 4, 6, 12, 13, 14, 19, 21)								$K = 85$	$A_{95} = 5.4$
Overall mean			$N = 9$	18	56	177	3.5	75	151
(site 5 added)								$K = 96$	$A_{95} = 4.8$
CRC method			$N = 8$	21	54	–	5.0	–	–

* Site: [a] omitted from computation of overall mean; [b] interpreted separately in the text. Age subdivisions are as in Table 1. n = Number of samples, a = studied, b = accepted, in brackets—number of specimens, when this level of statistics was used; N = number of sites. $D°$, $I°$ = mean declination and inclination, respectively. K = concentration parameter. α_{95} = radius of confidence circles. *Plat*, *Plong* = latitude and longitude of the pole, respectively.

(Halls, 1976). The first component was found to agree closely with the modern dipole field in the Kopet-Dagh: $D = 357°$, $I = 61°$ ($\alpha_{95} = 5°$) and $D = 0°$, $I = 58°$.

All of the Middle Cretaceous rocks revealed normal polarity, except for two reversely magnetized samples. Though these two vectors are nearly antiparallel to the normal ones, it is not enough for a conclusive reversal test.

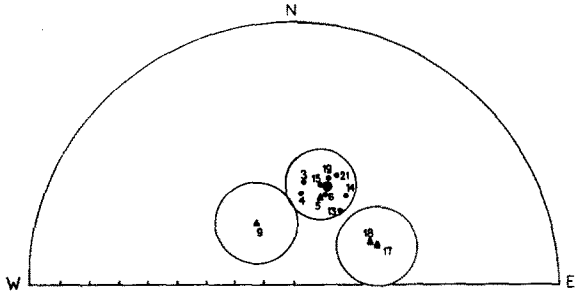


Fig. 4. Stereonet of Middle Cretaceous data after thermal cleaning and tilt correction. Legend: dots = site means of class A data; encircled dot = regional mean of class A data; triangles = other site mean directions (for explanations, see text). Site-mean confidence circles are shown only for results specially referred to in the text. Numbers correspond to numbers of sites in Fig. 1 and Table 2.

The fold test (McFadden and Jones, 1981) was applied to means computed after the final heating step. On a local scale, it is positive for site 12 and negative for sites 5, 18 and 20. On a regional scale, the test is positive for 6 sites (3, 6, 13, 14, 19 and 21) and negative for site 17. The CPR test (McElhinny, 1964) had to be used for site 4, where bedding attitudes varied widely. The concentration parameter ratio here is 3.4 and the critical value at a 95% confidence level is 1.6. Thus, the CPR test is clearly positive, and the fold test is also positive if this site mean is added to the other 6 "positive" sites. It can be concluded that the magnetization of Middle Cretaceous rocks at site 4 has a prefolding origin. The result from site 12 can not be directly compared with the other data because of the large disparity in concentration parameters (Table 2). But the confidence circle of

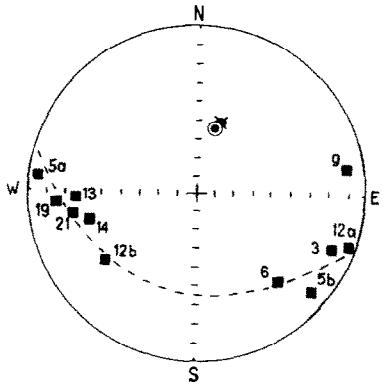


Fig. 5. Analysis of Middle Cretaceous data with the remagnetization circle method. Squares are site remagnetization circle normals. The dashed line is the great circle fitted to these normals (the normal for site 9 is excluded). The normal to this great circle is shown by the crossed dot. Encircled dot is the regional mean reproduced from Fig. 4. All data are presented in the unfolded state. Numbers correspond to numbers of sites in Fig. 1 and Tables 1 and 2.

this site mean overlaps all other "positive" means (Fig. 4), and from the author's point of view, these results can be safely combined. Thus, 8 tightly clustered site means (class A data) distributed over the Western and Central Kopet-Dagh can be used for the computation of a regional mean direction (Table 2).

As the next step, the CRC method (Halls, 1976) was used, first on a local and then on a regional scale. Subsites 5a and 5b give an intersection point which differs by only about 3° from the class A data mean. It would seem that this result could be used further. On the other hand, remagnetization circles for subsites 18a and 18b show quite a different intersection point, which falls well within the confidence circle for site 17 (Fig. 4). As will be shown below, such an agreement is not just a coincidence. Subsites 20a and 20b yielded no definite remagnetization circles and were both rejected. When the CRC method was used on regional scale (sites 17 and 18 were excluded), it yielded a direction lying only 2° apart from the overall mean (Table 2, Fig. 5).

The result from site 9 has not been discussed so far. This site mean is clearly anomalous (Table 2, Fig. 4), and its confidence circle does not overlap any other direction. The remagnetization circle deviates considerably as well (Fig. 5). Eight samples used for computation of the mean are distributed over a 300-m-thick pile of Cenomanian sandstones, and an anomaly in the geomagnetic field seems to be improbable. These sandstones are not separated by any angular unconformity or fault from the overlying Upper Cretaceous carbonates, which gave the "normal" result (see below). Thus, tectonics as a source of the paleomagnetic anomaly can be ruled out too. This result remained an enigma and was omitted.

The results of this Section can be summarized as follows. First, the regional mean can be based on either the class A data only, or the less reliable result from site 5 can be added. In principle, a postfolding component may still be present in these data sets in spite of a positive fold test, and that may result in a certain bias of the regional mean. Second, there is the CRC-defined direction, and this method has some inherent systematic errors (Schmidt, 1985). As it is, all three directions, the CRC-defined one and the two regional means, are in very close agreement (Table 2), and it is hardly possible that both the sources of systematic errors have resulted in similar biases. Thus, each of these three directions can be considered as an unbiased estimate of prefolding magnetization in the Middle Cretaceous rocks, and the class A data mean has been chosen for further analysis.

The prefolding nature of stable magnetization in the Middle Cretaceous rocks seems to be proved beyond doubt. Some evidence points to its primary origin: (1) the exclusively normal polarity agrees with the age of these rocks; (2) magnetite, detrital from optical observations, is the main NRM carrier; and (3) though the within-site scatter of paleomagnetic vectors varies rather widely (Table 2), the direction of prefolding magnetization does not depend upon this parameter. These facts are not conclusive, but on the other hand, there is no evidence of any prefolding remagnetization of these rocks.

Upper Cretaceous–Paleocene limestones and marls

The NRM intensities here range from $10–30 \cdot 10^{-6}$ emu in greenish-grey Coniacian–Santonian marls and redbeds to $0.2–0.4 \cdot 10^{-6}$ emu in light-

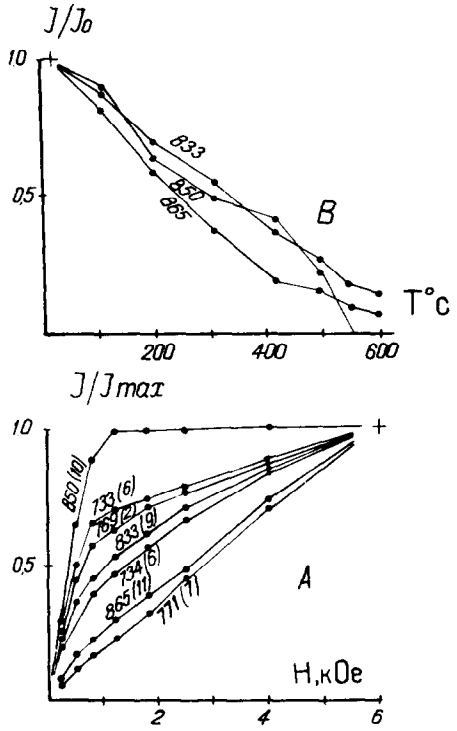


Fig. 6. Representative IRM acquisition (A) and IRM thermal demagnetization curves (B) for Upper Cretaceous–Paleocene rocks. Samples (sites) are numbered.

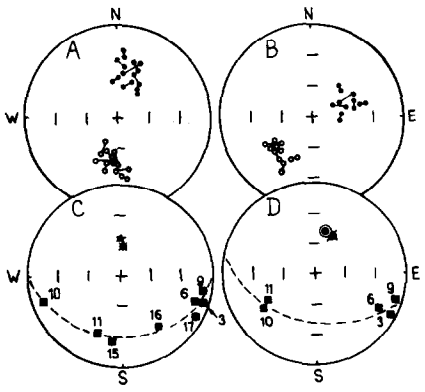


Fig. 7. Stereoplots of Upper Cretaceous–Paleocene data. A, B. Specimen direction distributions for sites 10 and 17, respectively. Specimen pairs from the same sample are linked (not all data are shown for clarity). C. Squares are site remagnetization circle normals. The dashed line is the great circle fitted to these normals. The normal to this great circle (crossed dot) is compared with the modern dipole field direction (star). All data are presented in geographic coordinates. Numbers correspond to numbers of sites in Fig. 1 and Tables 1 and 3. D. Squares and the dashed line are as above (C). The crossed dot denotes the normal to the best-fitting line. All data are tilt corrected. Numbers correspond to numbers of sites in Fig. 1 and Tables 1 and 3.

grey marls of Maastrichtian and Paleocene age. The rock-magnetic studies showed that magnetite and haematite in varying proportions were the NRM carriers (Fig. 6). A part of the collection, white limestones mostly, proved to be stable above 200 °C. However, the main body of samples showed regular shifts of paleomagnetic directions

TABLE 3

Upper Cretaceous–Paleocene paleomagnetic results

Site	Age	n		$D°$	$I°$	K	α_{95}	*Plat* (°N)	*Plong* (°E)
		a	b						
1	m–Tp	40	26	16	46	14	7.4	73	181
3	st–Tp	13	8	18	43	12	14.4	70	184
6	t–m	43	36	10	45	13	6.5	75	199
9	t–Tp	39	31	8	45	23	5.2	76	205
10	cn–Tp	46	35	8	47	20	5.4	77	202
11	cn–Tp	38	26	15	45	29	5.1	72	186
15 [a]	cp–Tp	18	11	36	41	61	5.2	57	166
16 [a]	cp–Tp	20	14	57	51	43	5.7	44	156
17 [a]	cn–m	17	15	54	46	54	5.2	44	144
M	Tp	23	20	10	41	7	12	75	204
Overall mean (7 sites)				12	45	526	2.6	74 $K = 460$	193 $A_{95} = 2.8$
Limestones (sites 6, 9, 10, 11)				12	51	350	4.0	78 $K = 250$	179 $A_{95} = 4.5$
CRC method (sites 3, 6, 9, 10, 11)				10	46	–	3	–	–

[a] Interpreted separately in the text. See Table 2 for explanations of other symbols.

within the whole temperature range, thus pointing to incomplete elimination of unstable NRM components. As a rule, the within-sample grouping of vectors was much better than the between-sample grouping (Fig. 7A, B). Both normally and reversely magnetized samples were found at most sites.

The site-means are of different reliability. Some are obtained for large collections with both polarities present, whereas others are based on few unipolar samples. Those results where the direction of bipolar magnetization have been obtained with the aid of the reversal test fall into two groups. The minor group includes sites 15–17, with clear northeastern declinations, from a limited area in the southwestern part of the Kopet-Dagh. These directions are rather similar to those measured in the Middle Cretaceous rocks (sites 17 and 18, Table 2) and their significance will be discussed in a separate section. The major group includes six sites (1, 3, 6, 9, 10 and 11) from all over the Kopet-Dagh. These site means (plus M site) are very tightly clustered and clearly pass the fold test. Only they were used for calculation of the overall mean direction (Table 3). Rocks of the same age were also studied at sites 5, 7 and 13, where only vectors of one polarity were found. These site means (not presented here), based on three to six samples, are poorly defined, with α_{95} up to 20°, and do not pass the fold test. As the result, they were rejected.

The sites where both polarities were recovered have been analyzed in another way as well. It is evident that if the mean R and N directions from a site are not antipodal, a unique great circle can be drawn through these two vectors. Assuming a bicomponent character of magnetization, an intersection of several circles will define both postfolding and prefolding remanences in the folded and unfolded states, respectively, provided there is an appreciable between-site structural diversity. It was found that the postfolding component ($D = 2°$, $I = 60°$, $\alpha_{95} = 5°$) agreed with the modern dipole field ($D = 0$, $I = 58°$). After heating to 200°C and above, the R and N means from site 1 are perfectly antiparallel and thus could not be used.

Two points should be stressed here: (1) normals to remagnetization circles closely fit the same plane (Fig. 7C) irrespective of direction of prefolding remanence; and (2) the postfolding components in the Middle Cretaceous sandstones and the Upper Cretaceous–Paleocene limestones and marls are very close to each other and to the modern dipole field. Thus, it can be concluded that all over the Kopet-Dagh, the sedimentary rocks bear the same recent post-tectonic component, probably related to weathering processes.

After tilt corrections, the CRC-defined and the overalll mean directions closely match each other (Table 3, Fig. 7D). Using the same reasoning as for the Middle Cretaceous data, it can be stated that these two results are to be considered as an unbiased estimate of prefolding magnetization.

Some evidence points to the primary age of stable magnetization in these rocks. First there is the very presence of both the N and R directions, which are almost antipodal sometimes (Fig. 7A, B), and the lateral traceability of polarity zones. Although the sampling density was in general inadequate for accurate between-site zone matching, it could be done for some marker horizons. For example, the Lower Campanian white limestones (the upper part of the beds with *Offaster pomeli* Mun.-Ch.; Atabekyan and Likhacheva, 1961), some 25–40-m thick, are reversely magnetized over 200 km (sites 5, 6, 11 and 17).

Second, the within-sample scatter of paleomagnetic vectors is generally much lower than the between-sample scatter (Fig. 7A, B); in other words, sample means are statistically distinct. Such a pattern may result from a partial remagnetization if directions of primary and secondary components are widely divergent and their ratio changes from sample to sample. However, this would most probably lead to a smeared unit-vector distribution, which is clearly not the case here. Any complete remagnetization would result in uniform remanence of large volumes of rocks and not in the observed relation of the within-sample and between-sample dispersals. The only solution seems to be a syndepositional process of NRM acquisition where each thin layer is magnetized along the instantaneous direction of the ambient field, and sample-mean scatter is due to secular variations.

Thus, two independent lines of evidence point

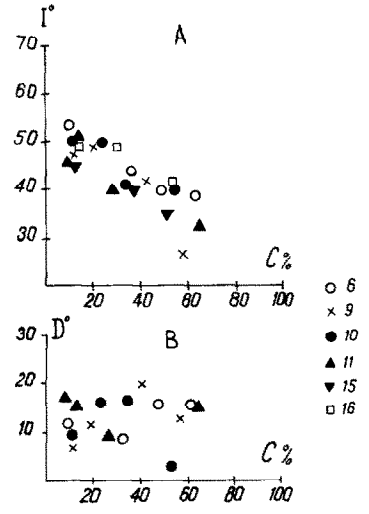


Fig. 8. Diagrams of inclination (A) and declination (B) versus the quantity of insoluble material (C%) for Upper Cretaceous–Paleocene rocks. Numbers correspond to the site numbers in Table 1.

to the primary origin of the bipolar magnetization in the Upper Cretaceous–Paleocene rocks; more evidence will be presented below. Of course, the above conclusion would preserve its validity if this magnetization was "contaminated" by some weak, secondary prefolding NRM component, but its direction has been "purified" with the aid of the reversal test.

Some more data should be presented. At face value, the site-mean inclinations (Table 3) appear to be very homogenous, all differences being statistically insignificant. But this consistency is deceiving. The percent of material (C) insoluble in weak hydrochloric acid was determined for each sample from sites 6, 9 10, 11, 15 and 16. This parameter, C, is closely related to clay content in the sedimentary rocks under study. Each collection was subdivided into three or four groups according to C values in such a way as to make the reversal test applicable for the calculation of each group mean. Thus, obtained group-mean inclinations correlate strongly (correlation coefficient $r = -0.85$) with parameter C (Fig. 8A), while declinations do not ($r = 0.10$, Fig. 8B). The anomalous declination for sites 15 and 16 were excluded from the calculations.

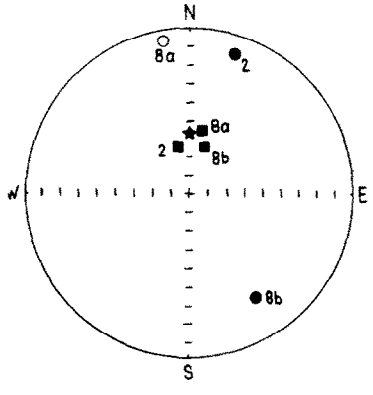


Fig. 9. Stereonet of Eocene–Oligocene data. The squares (dots) denote site means before (after) tilt correction. The star is the modern dipole field direction.

The Eocene–Oligocene clays

These rocks were studied at sites 2 and 8; some 5–7 samples were taken at a number of other sites too. The NRM intensities are usually low, about $1 \cdot 10^{-6}$ emu, and often drop to the noise level of the magnetometer when heated above 300°C. After thermal cleaning at 250°C, site means are widely divergent in stratigraphic coordinates, and closely match each other and the modern dipole field direction in geographic coordinates (Fig. 9). Thus, the magnetization of Eocene–Oligocene rocks is clearly of recent postfolding origin, and these data can not be used further.

Interpretation and discussion

The correlation between lithology and paleomagnetic inclinations

Two explanations for the inclination error have been proposed: a model based on shape anisotropy of magnetic particles (King, 1955) and another one based on rolling spheres (Griffiths et al., 1960). Both require the falling of rather large grains upon the rigid surface of earlier-deposited sediment. All of the Upper Cretaceous and Paleocene rocks are very fine-grained, and had been water-saturated oozes before lithification. Thus, neither model can account for the above-described correlation (Fig. 8). Besides, they do not explain why inclinations vary widely in sediments with similar grain size distributions.

Different types of rocks are interlayered within

the Upper Cretaceous–Paleocene sequence. The inclination values decrease rather steadily from limestones ($C = 0$–20%) to marls (C = about 60%) (Fig. 8A). Moreover, lateral changes in lithology are reflected in inclinations. For example, the Coniacian–Santonian rocks at sites 9, 10 and 11 are more carbonatitic than the rocks at site 6; the corresponding inclinations are $I = 49 \pm 3°$ and $I = 34 \pm 5°$. These facts preclude tectonics as a source of correlation.

It was assumed that data observed might be explained by diagenetic compaction of sediments. Such a hypothesis has already been proposed (e.g., Laj et al., 1982), although without numerical evaluation. If correct, this hypothesis leads to some important conclusions:

(1) As compaction of sediments takes place mainly at early stages of diagenesis, such a correlation implies that magnetization must be either contemporaneous with sedimentation in case of DRM or quasicontemporaneous with it in case of post-depositional DRM and early diagenetic CRM. Thus, the strong correlation between inclination and clay content confirms the primary origin of stable magnetization in the Upper Cretaceous–Paleocene rocks and, in general, such an approach may be used for component dating.

(2) Only the largest values of inclination should be used for paleolatitude estimation. In this paper, these are the data from limestones at sites 6, 9, 10 and 11. The overall mean for all types of rocks studied will lead to errors. The steepest inclinations also may be shallow, but such a selection will result in minimal errors.

(3) Paleomagnetism can be used, in principle, as a tool for investigation of compaction processes in sediments.

Local tectonics

The site-mean directions for both the Middle Cretaceous and Upper Cretaceous–Paleocene rocks (Tables 3 and 4) are very consistent across the Kopet-Dagh, except for a limited area in the southwestern part. The northeastern declinations at sites 15–18 (Fig. 10A) are in sharp contrast with the other data. Both the paleomagnetic and rock-magnetic characteristics at anomalous sites are similar to those observed everywhere and there is no reason to believe that this deflection is due to some remagnetization. Thus, tectonics seems to be the only explanation, and only the mode of movements is to be discussed.

Rigid-body postfolding rotation of a block some tens of kilometers long will lead to overlapping of folds and/or a very complex structural pattern *before rotation*; this is highly improbable. Rigid-body prefolding rotation of the same block implies considerable thrusting and/or other types of deformation older than the main folding. No geo-

TABLE 4

Pole positions and paleolatitudes for extra-Alpine Eurasia and the Kopet-Dagh (reference point at 39 °N, 56 °E) *

Region and age	Symbol	*Plat* (°N)	*Plong* (°E)	K	α_{95}	*pl* °	References
Eurasia, K_1	E1	69	170	290	3.1	27 ± 3	(1)
Eurasia, K_1	E2	73	154	230	3.7	34 ± 4	(1)
Eurasia, K_u	EK	74	152	91	13	–	(2)
Eurasia, T_{p-e}	EP	76	161	165	3	33 ± 3	(2)
Kopet-Dagh, K_m	KM	75	153	85	5.4	36 ± 5	(3)
Kopet-Dagh, K_u–T_p	KU	74	193	460	2.8	25 ± 3	(3)
Kopet-Dagh, K_u, limestones only	L	78	179	250	4.5	31 ± 4	(3)

* *Plat, Plong* = latitude and longitude of the paleopole, respectively. K = concentration parameter. α_{95} = radius of confidence circle. *pl* = latitude of reference point together with its error limits. References: (1) = Bazhenov and Shipunov (1985); (2) = Westphal et al. (1986); (3) = this paper.

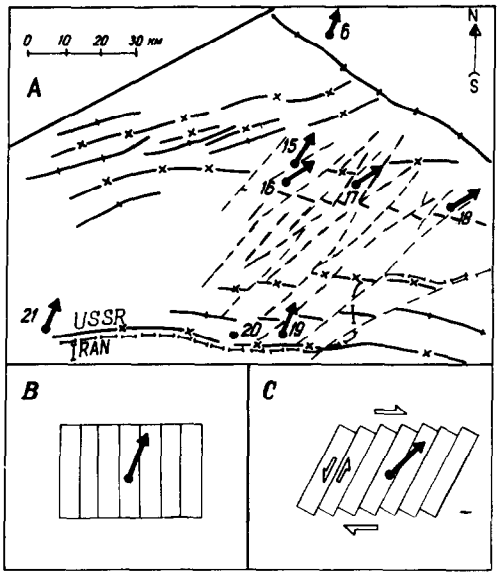


Fig. 10. Interpretation of paleomagnetic data from the southwestern part of the Kopet-Dagh. A. Fold-axis pattern (after Rastsvetaev, 1972, simplified) and site-mean declinations. Thin dashed lines denote mapped strike-slip faults. Fold axes (thick lines) are copied from Fig. 1. B, C. Inferred model of movements. Solid arrows are paleomagnetic vectors, open arrows are directions of tectonic stress.

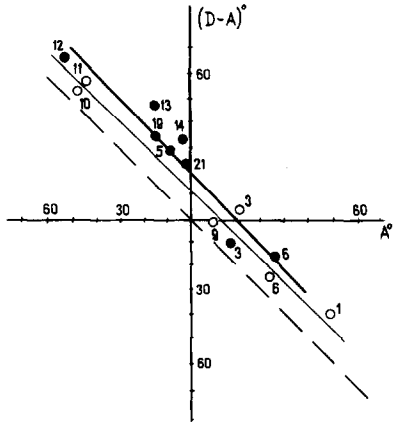


Fig. 11. Data analysis by regression technique. The thick and thin solid lines are the best-fitting lines for the Middle Cretaceous (full dots) and the Upper Cretaceous–Paleocene (open dots), respectively; the dashed line is the theoretical plot for a primary arc.

logical manifestation of such movements is known so far. A plunging of fold axes could lead to declination anomalies, but of the opposite signs at both limbs of a fold (Rzhevsky and Slautsutays, 1976). In this case, all of the site means, from the limbs of one fold in particular (i.e., sites 15 and 16), point to clockwise rotation and agree rather well with each other over this area. Moreover, all of the fold axes here are subhorizontal. Thus, this mechanism can be rejected as well.

The key fact is the system of parallel, sinistral, low-magnitude strike-slip faults developed in the area. The mode of rotation proposed is clear from Figs. 10B and C. This model does not require any distortion of the fold-axes pattern and explains both paleomagnetic and structural data.

Very similar models have already been proposed for the tectonic interpretation of paleomagnetic results (Freund, 1970; Fagin and Gose, 1983). However, both the block rotations and the system of subparallel strike-slips were usually considered to be the result of movements along some large strike-slip fault. There are no geological manifestations of east–west-trending large faults in this part of the Kopet-Dagh. It is possible to assume that this area represents an initial stage of the typical wrench-zone evolution. According to geometry and paleomagnetic data, the displacement between the southern and northern flanks of the faulted area did not exceed some tens of kilometers.

Another problem is the origin of the arc-like structural pattern in the western part of the Kopet-Dagh. It has already been concluded (Bazhenov, 1979) that these bends are primary. This statement can be checked more accurately with functions like $D° = f(A°)$ or $(D-A)° = f(A°)$ where D is the site-mean paleomagnetic declination and A is the mean azimuth of strike (or dip) at a sampling locality (Ries at al., 1980; Bazhenov, 1982). It is easy to show that for a primary arc-like structure, the plot $(D-A)° = f(A°)$ will be a straight line inclined at an angle $m = -45°$ to the $A°$ axis. The numerical values for the analysis were taken from Tables 1, 3 and 4. The plots for the Middle Cretaceous and Upper Cretaceous–Paleocene data sets separately agree quite well with the theoretical line, with $m = -45°$ (Fig. 11). This result, as well as the fact that no large thrusts are known in the Kopet-Dagh so far, leads to the conclusion that the bends of the fold axes are primary and resulted from differently oriented stresses (Rastsvetaev, 1972).

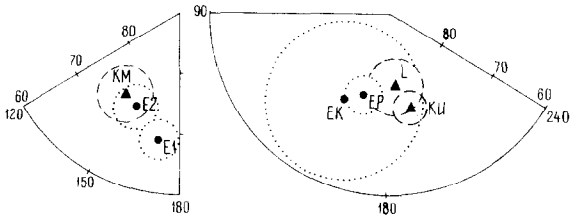


Fig. 12. Paleomagnetic poles for the Kopet-Dagh and extra-Alpine Eurasia (all data are shown on the upper hemisphere of the modern North pole-centered projection). Dashed (dotted) lines denote confidence circles for the Kopet-Dagh (Eurasia) poles (poles are labelled as in Table 4 and the text).

Regional tectonics

In order to determine the movement of the Kopet-Dagh as a whole, some reference poles for extra-Alpine Eurasia were taken. The Early–Middle Cretaceous pole (including some new data) has been calculated in two ways: as the usual mean and as the intersection of paleomeridians (Bazhenov and Shipunov, 1985) (poles E1 and E2, Table 4, Fig. 12). The difference between these two estimates, however small, is statistically significant. Though both may be used, the latter result was shown to be the more reliable estimate of the true Early–Middle Cretaceous paleomagnetic pole. As for the Upper Cretaceous and Paleogene (T_{p-e}) poles (Westphal et al., 1986), the first one (EK) is of no help here owing to its very low accuracy ($A_{95} = 13°$). In contrast, the Paleogene reference pole (EP, Table 4) is precisely defined and can be used here, as the pole shift for the Cretaceous–Paleogene time interval is so small that no estimable error may stem from substituting the EP pole in place of the poorly defined Upper Cretaceous pole.

The Middle Cretaceous position of the Kopet-Dagh (KM pole) agrees quite well with the paleolatitude, recalculated from the Eurasian data (E2 pole). (Note that the E1 pole will lead to an overlapping of the Kopet-Dagh onto the Turan plate, which is impossible from a geological point of view). The mean Upper Cretaceous–Paleocene latitude is considerably lower than the predicted one, while the result for limestones only (L pole) agrees well with the Eurasian grid. This is not surprising in view of the previously described flattening of inclinations in a part of the Upper Cretaceous rocks. Therefore, one can come to the conclusion that the Kopet-Dagh as a whole did not move within the error limits in relation to extra-Alpine Eurasia. Though the author failed to determine accurately these limits for the whole data set, the regional movement, if any, hardly exceeded 200 km.

A brief overview of the Cretaceous and Cenozoic paleomagnetic results from the outermost parts of the Alpine belt shows that there are some other stable domains. For example, the Great Caucasus appears to have been welded to the Eurasian plate, according to Early Cretaceous data (Tretyak et al., 1976). Horizontal movements within such domains were either restricted to rotations of minor structures or absent. In contrast, other marginal parts of the belt have suffered considerable movement with respect to extra-Alpine Eurasia. For instance, the external tectonic zones of the Pamirs were translated northwards about 400 km and, moreover, were severely deformed (Rzhevsky, 1977; Bazhenov and Burtman, 1982). Thus, two types of tectonics can be recognized in the marginal parts of the Alpine belt. However, at present, one should abstain from further conclusions because of the scarcity of data.

Conclusions

The main results of the article run as follows:

(1) The compaction of sedimentary rocks can affect paleomagnetic inclinations, leaving declinations unchanged. This can be used for magnetization dating, and should be accounted for in methodology, especially paleolatitude estimation.

(2) The structural pattern in the Kopet-Dagh is primary, except for minor distortions caused by rotations within the local wrench zone.

(3) Within error limits, the Kopet-Dagh as a whole did not move in relation to extra-Alpine Eurasia.

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